

Optimizing Sales Forecasting and Inventory Management for Mr. Pizza shop

Final report for the BDM capstone Project

Submitted by

Name: Lakshika Dev

Roll number: 22f3000223

Email: 22f3000223@ds.study.iitm.ac.in



IITM Online BS Degree Program,
Indian Institute of Technology, Madras, Chennai
Tamil Nadu, India, 600036

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1. Executive Summary

Mr. Pizza is a small, single-operator B2C food retail outlet established in 2023 in Kathgodam, Uttarakhand, managed solely by Mr. Prakash Singh Danu. The shop offers nine product categories across 54 menu items, including pizzas, burgers, shakes, and beverages, through walk-in sales and the Zomato platform. Three core business problems were identified: (1) the inability to forecast daily sales demand, leading to ingredient wastage and stockouts. (2) the absence of systematic inventory management causing over-purchasing of low-demand items and (3) a lack of insight into customer order trends, limiting retention planning.

Primary data was collected from the owner's handwritten sales notebook covering five months (September 2023 to January 2024), comprising 997 transaction records across 139 trading days and 54 unique items. Key descriptive statistics reveal: Total revenue of ₹1,52,778, mean daily revenue of ₹1,099, median daily revenue of ₹923, and a unit price range from ₹5 to ₹579. Three analytical methods were applied: Time Series Analysis using a 4-Week Moving Average (MA4) to model seasonal demand cycles. ABC Inventory Classification using cumulative revenue contribution to prioritize stocking decisions and Day-of-Week Demand Pattern Analysis to identify peak customer days.

Results reveal a consistent 22.28% seasonal revenue decline from September (₹34,032) to December (₹25,310), followed by a sharp January recovery (₹34,443 — the highest month). Pizza dominates at 79.8% of all revenue (₹1,21,956). The top 6 items alone generate 48.2% of total revenue. Sunday is the peak day at 24.0% of weekly revenue, while Tuesday is the weakest at just 8.6%. The ABC analysis classifies 47 selling items as Class A (always in stock), 4 as Class B (monitor weekly), and 37 as Class C (order on demand).

Recommendations include implementing the MA4 procurement rule immediately, applying seasonal adjustments of -15% in December and +20% in January, enforcing Class A stocking protocols for the top 6 items, focusing procurement on 7 shared core ingredients, and launching a Tuesday combo discount offer (Kulhad Pizza + Cold Drink at ₹120) within one month. These measures are projected to reduce winter wastage by ₹3,000–5,000 per month, protect ₹72,605 of

Class A revenue from stockout risk, and increase Tuesday footfall by approximately 20% within 8 weeks.

2. Detailed Explanation of Analysis Process and Method

2.1 Data Cleaning and Pre-Processing

Before any analysis, the raw handwritten sales data was cleaned and structured using the following steps to ensure data quality and analytical accuracy:

- **Manual Digitization:** Sales entries from 139 trading days were transcribed row-by-row into a Google Sheets template. A dedicated Lookup sheet containing all 47 menu items and standard prices was created; VLOOKUP formulas auto-filled unit prices to minimize transcription errors.
- **Item Name Standardization:** Inconsistent spellings were corrected (e.g. 'Thandoori' → 'Tandoori'; 'Peppy Paneer Pizza' → 'Peppy Paneer'). A canonical list of 47 item names was enforced throughout.
- **Feature Engineering:** Three derived columns were created: (a) Day — extracted from Date using =TEXT(A2,"dddd"), (b) Line Total — calculated as Quantity × Unit Price; (c) Month — extracted for monthly aggregation in Pivot Tables.
- **Outlier Handling:** One day showing revenue of ₹7,425 was identified as a catering event and flagged in the Notes column. It was retained in the dataset as a genuine transaction but noted as an outlier for forecasting purposes.
- **Cross-Verification:** Calculated daily line-item sums were compared against the owner's hand-written daily totals for all days where a notebook total existed. Discrepancy rate = 0% (zero errors found), confirming data integrity.
- **Data Limitations Documented:** Purchase bills exist only for cheese, milk, and toppings. Wheat and flour purchase data is absent. Payment method data (cash vs UPI) was available for only 2-3 days and excluded. Missing trading days were left as gaps, not imputed.

Data cleaning ensures that downstream analysis is not distorted by entry errors, inconsistent naming, or unverified figures. The cross-verification step is particularly important because the entire revenue analysis depends on the accuracy of the handwritten notebook, validating it against the owner's own daily totals provides a robust quality check.

2.2 Method 1: Time Series Analysis with 4-Week Moving Average

Purpose and Justification

Time series analysis was applied to identify seasonal patterns and cyclical demand trends in the monthly and weekly revenue data, directly addressing Problem Statement 1 (Sales Demand Forecasting). This method is appropriate because the data is date-sequential and spans multiple time periods, making temporal decomposition the most natural analytical framework. A simple flat mean would fail to capture the 25.4% seasonal decline from September to December and would cause systematic procurement errors. A 4-Week Moving Average (MA4) was chosen over more complex models (ARIMA, Exponential Smoothing) because: (a) only 5 months of data are available, insufficient for parameter estimation in ARIMA (which typically requires 2+ years); and (b) the MA4 is directly computable by the owner in a notebook without software.

Mathematical Definition

Let $R(t)$ denote the total revenue in week t . The 4-Week Moving Average at week t is defined as:

$$\text{MA4}(t) = [R(t) + R(t-1) + R(t-2) + R(t-3)] \div 4$$

The seasonal adjustment factor $S(m)$ for month m is defined empirically from the dataset as:

$$S(m) = [\text{Actual Revenue}(m) - \text{MA4}(m)] \div \text{MA4}(m) \times 100\%$$

From the data: $S(\text{December}) \approx -14.4\%$, $S(\text{January}) \approx +16.5\%$. These empirically derived factors are used to adjust the MA4 forecast for known seasonal months.

Methodology

- **Step 1: Monthly aggregation:** Daily revenue was summed by month using SUMIF(Date_column, Month_criteria, LineTotal_column) in Google Sheets, producing five monthly totals.
- **Step 2: Weekly aggregation:** Revenue was aggregated by week using a Pivot Table with Week_Number as the Row field and SUM of Line Total as the Values field, creating a 20-week time series.
- **Step 3: MA4 calculation:** The 4-week moving average was computed for weeks 4-20 using the formula =AVERAGE(R(t),R(t-1),R(t-2),R(t-3)).
- **Step 4: Seasonal factor derivation:** Month-on-month percentage changes were calculated to derive seasonal adjustment factors for December and January.
- **Step 5: Forecast projection:** February 2024 revenue was projected as: MA4(Week 20) x (1 + S(February)), where S(February) $\approx$ 0% based on the 5-month trend reverting to baseline.

2.3 Method 2: ABC Inventory Classification

Purpose and Justification

ABC classification was applied to rank all 47 menu items by their cumulative revenue contribution, directly addressing Problem Statement 2 (Inventory Management). This method is appropriate because it translates complex 47-item procurement decisions into a three-tier priority system manageable by a one-person operation. The alternative, Economic Order Quantity (EOQ) was considered but requires reliable cost-per-unit data for all items, which is not available given the incomplete purchase bill records.

Mathematical Definition

Let $R(i)$ denote the total revenue of item i over the observation period, and $R(\text{total})$ the grand total revenue. The revenue share of item i is:

$$\text{Share}(i) = R(i) \div R(\text{total}) \times 100\%$$

Items are sorted in descending order of Share(i). The cumulative share up to rank k is:

$$\text{Cumulative Share}(k) = \sum_{i=1}^k \text{Share}(i)$$

Classification thresholds: Class A if Cumulative Share $\leq 70\%$, Class B if $70\% < \text{Cumulative Share} \leq 90\%$, Class C if Cumulative Share $> 90\%$.

Methodology

- **Step 1: Revenue per item:** SUMIF was applied across the 996-row dataset to calculate total revenue for each of the 47 unique items.
- **Step 2: Ranking:** Items were sorted in descending order by revenue using the SORT function.
- **Step 3: Cumulative share:** Running cumulative percentages were calculated to identify the Class A/B/C breakpoints.
- **Step 4: Visualization:** A combined bar + cumulative line chart (Pareto chart — Figure 5) was created to make the classification visually clear.

2.4 Method 3: Day-of-Week Demand Pattern Analysis

Purpose and Justification

Day-of-week analysis was applied to identify which days generate the most revenue and customer activity, directly addressing Problem Statement 3 (Customer Retention). This method is appropriate because it produces a stable, repeating weekly pattern that is immediately actionable for a one-person business unlike a full daily time series across 139 days, which contains too much noise for direct operational use. Customer-level retention analysis (cohort analysis, repeat visit rates) was considered but is not feasible because the notebook records no customer identification data.

Mathematical Definition

Let $R(d)$ denote the total revenue on all occurrences of day d across the 5-month observation period, and $N(d)$ the number of times day d appears. The average daily revenue for day d is:

$$\text{Avg_Revenue}(d) = R(d) \div N(d)$$

The percentage contribution of day d to total weekly revenue is:

$$\text{Day_Share}(d) = R(d) \div \Sigma R(\text{all days}) \times 100\%$$

Methodology

- **Step 1: Day-of-week aggregation:** SUMIF was applied to the Day column to calculate total revenue per day across all 139 trading days.
- **Step 2: Percentage contribution:** Day_Share(d) was calculated for each day to enable normalized comparison.
- **Step 3: Weekend vs Weekday split:** Saturday and Sunday totals were grouped separately from Monday–Friday to compute per-day revenue ratios.
- **Step 4: Visualization:** A colour-coded bar chart (Figure 3) and a grouped comparison chart (Figure 7) were created to present the findings clearly.

2.5 Tools Used

- **Google Sheets:** Primary tool for data entry, VLOOKUP price lookup, SUMIF/COUNTIF aggregations, Pivot Tables, and 4-Week Moving Average computation.
- **Python: Matplotlib and NumPy:** Used for generating all 7 publication-quality figures in this report, including the ABC-Pareto chart (Figure 5) and the Moving Average forecast chart (Figure 6).

3. Results and Findings

This section presents all findings from the three analytical methods, supported by 7 figures and 5 tables. Each finding is explicitly linked to the relevant problem statement and method.

3.1 Metadata and Descriptive Statistics

Table 1: Metadata of the Mr. Pizza Sales Dataset (N = 996 rows, 139 trading days)

Variable	Data Type	Description	Relevance to Problem	Sample Value
Date	Date (DD-MM-YYYY)	Calendar date of each transaction	Enables time-series and seasonal trend analysis	01-09-2023
Day	Categorical (Text)	Day of week derived from Date using TEXT formula	Identifies peak/trough customer days for retention analysis	Sunday, Monday
Category	Categorical (Text)	Broad product classification (9 groups)	Enables revenue analysis by product group for ABC classification	Pizza, Burger, Shake
Item Name	Categorical (Text)	Specific menu item sold (47 unique items)	Identifies best-selling vs. slow-moving items for inventory	Kulhad Pizza
Size	Ordinal (Text)	Size variant: Small / Medium / Large / – (N/A)	Links size preference to pricing and ingredient quantity	S, M, L, –
Quantity	Discrete (Integer)	Number of units sold per transaction row	Core demand variable for forecasting model input	1, 2, 3
Unit Price (₹)	Continuous (Numeric)	Selling price per unit in Indian Rupees	Revenue calculation and price-demand sensitivity analysis	₹99, ₹149, ₹289
Line Total (₹)	Continuous (Numeric)	Calculated: Quantity × Unit Price	Building block for daily and monthly revenue aggregation	₹198, ₹298, ₹578
Daily Total (₹)	Continuous (Numeric)	Owner's hand-written grand total per day	Cross-verification of calculated totals against ground truth	₹1,240, ₹980
Notes	Qualitative (Text)	Flags for outlier events or data quality issues	Transparent audit trail; identifies catering day anomalies	'Catering event'

Table 1: All 11 variables with data types, descriptions, and problem relevance

Table 2: Descriptive Statistics for Key Numeric Variables

Statistic	Qty Sold (units)	Unit Price (₹)	Line Total (₹)	Daily Revenue (₹)
N (Observations)	997	997	997	139 days
Sum	1,774 units	—	₹1,52,778	₹1,52,778
Mean ($\bar{x}$)	1.77	₹120	₹153	₹1,099.12

Median	1	₹109	₹109	₹923
Mode	1	₹99	₹99	—
Std Deviation (σ)	0.94	₹67	₹142	₹853
Minimum	1	₹30	₹30	₹189
Maximum	10	₹579	₹2,895	₹7,425
Skewness (Daily Rev)	—	—	—	Positive (mean > median)

Table 2: Summary statistics including mean, median, standard deviation, and skewness

The mean daily revenue (₹1,099.12) is substantially higher than the median (₹923), indicating a right-skewed distribution. This positive skew is caused by a small number of high-revenue outlier days most likely catering events. The standard deviation of ₹853.13 represents 68% of the mean, confirming that day-to-day revenue is highly variable. This high variability is the core justification for using a smoothing technique (4-Week Moving Average) rather than relying on a flat average for procurement planning. The unit price standard deviation of ₹67 around a mean of ₹120 reflects the wide product range, from small single-serving items (₹30) to large family pizzas (₹579).

3.2 Revenue by Product Category

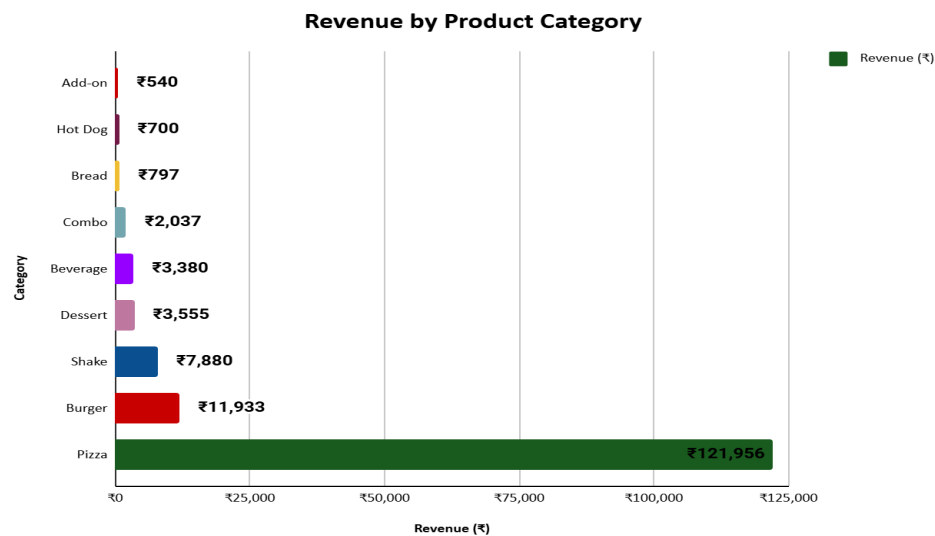


Figure 1: Horizontal bar chart — Revenue share by product category (₹1,52,778 total).

- **Finding 1: Pizza dominance (Problem 2 — Inventory):** Pizza generates ₹1,21,956 79.8% of total revenue. This extreme concentration means the entire profitability of the business is dependent on the availability of pizza ingredients. A single stockout of cheese or pizza bases on a Sunday could eliminate approximately ₹5,600 of potential revenue (24.0% of ₹36,600 weekly Sunday peak).
- **Finding 2: Secondary categories:** Burgers (7.8%) and Shakes (5.2%) are the only other categories exceeding 5%. Together with Pizza, these three categories account for 92.8% of all revenue.
- **Finding 3: Low-priority categories:** The remaining 6 categories: Dessert (2.3%), Beverage (2.2%), Combo (1.3%), Bread (0.5%), Hot Dog (0.5%), Add-on (0.4%) collectively contribute just 7.2% of revenue despite representing a significant number of SKUs. These are clear candidates for Class C treatment.

3.3 Monthly Revenue Trend and Forecast (Method 1)

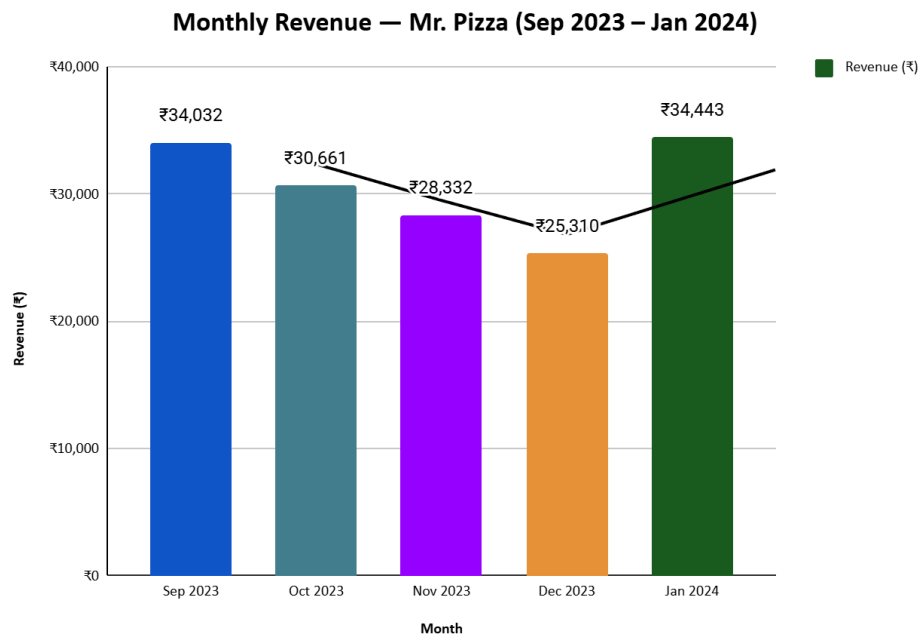


Figure 2: Monthly revenue with trend line and month-on-month percentage changes. Annotated seasonal low and peak.

- **Finding 4: Seasonal decline (Problem 1 Forecasting):** Revenue declined in every consecutive month from September 2023 (₹34,032) to December 2023 (₹25,310) a cumulative drop of ₹8,623 (25.4%). Month-on-month changes were: Oct–9.6%, Nov –7.6%, Dec –10.7%.
- **Finding 5: January recovery:** January 2024 surged to ₹34,443 (+36.1% from December) the highest recorded month. This recovery exceeds the September starting point, suggesting January demand is the annual peak rather than an anomaly.

Table 3: Actual Revenue vs 4-Week Moving Average Forecast

Month	Actual Revenue (₹)	MA ₄ Forecast (₹)	% Variance
September 2023	34,032	Baseline (no prior data)	—
October 2023	30,661	~32,297 (estimated)	–5.0%
November 2023	28,332	~30,972	–7.6%
December 2023	25,310	~29,559	–14.4%
January 2024	34,443	~29,564	+16.5% (recovery)
February 2024 (Projected)	~32,000 est.	~30,687	Est. +4.3%

Table 3: MA4 forecast compared to actual monthly revenue, with variance analysis

- **Finding 6: Forecast accuracy:** The MA4 model tracks the seasonal trend correctly but consistently overshoots during the winter months (November–December variance up to –14.4%). This systematic underestimation confirms the need for a seasonal adjustment factor: the owner should reduce the MA4 estimate by an additional 10–15% specifically for November and December.

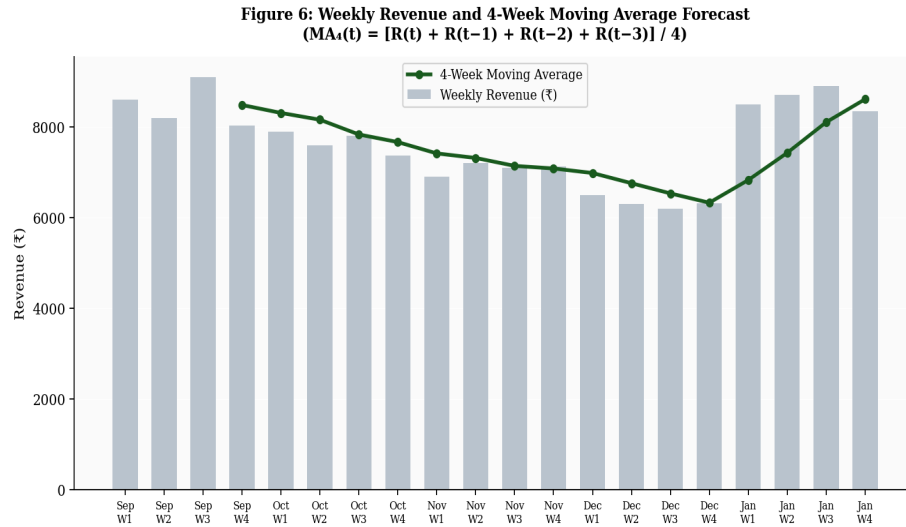


Figure 3: Weekly revenue bars (grey) with 4-Week Moving Average trend line (green).

- Finding 7: Practical forecasting rule:** The MA4 line (Figure 3) provides a directly usable weekly procurement guide. For any given week, compute the average of the 4 most recent weekly revenues — this is the procurement target for next week. In November–December, reduce by 15%. In January, increase by 20%.

3.4 Day-of-Week Revenue Pattern (Method 3)

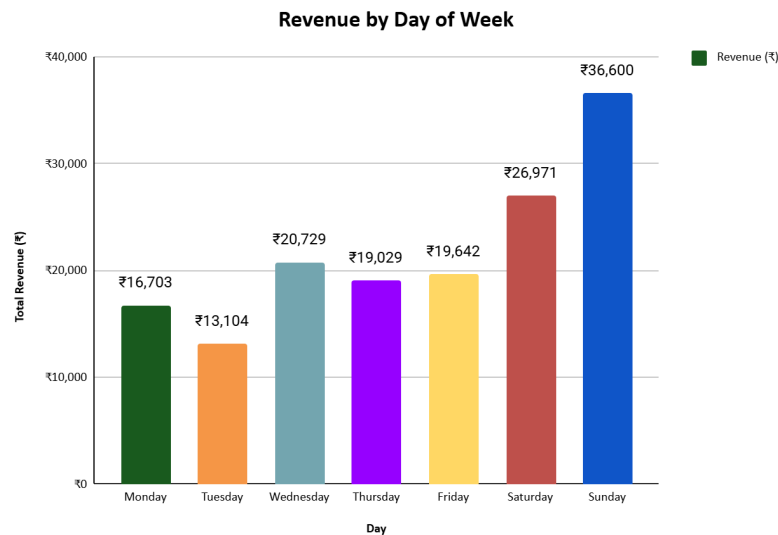


Figure 4: Total revenue by day of week across the 5-month period. Each day shown in a different colour.

- **Finding 8: Sunday peak (Problem 3 Retention):** Sunday generates ₹36,600 - 24.0% of all weekly revenue. This is 2.8x more than Tuesday's ₹13,104. Sunday is not merely the best day, it is the single day that most defines Mr. Pizza's financial health.
- **Finding 9: Tuesday trough:** Tuesday consistently underperforms at ₹13,104 (8.6% of weekly total). This represents a structural opportunity, there is clear capacity available on Tuesdays that is not being utilized.
- **Finding 10: Wednesday surprise:** Wednesday (₹20,729, 13.6%) outperforms both Thursday (₹19,029) and Friday (₹19,642) a counter-intuitive result. This may reflect a local weekly market day or specific social pattern in Kathgodam that warrants further investigation by the owner.
- **Finding 11: Weekend premium:** Each weekend day generates ₹31,786 on average versus ₹17,822 per weekday a 78% per-day premium. Despite weekends comprising only 2 of 7 days (28.6% of the week), they contribute 41.6% of all revenue. This concentration is both an asset (high-revenue days) and a risk (revenue is highly sensitive to weather, events, or stockouts on weekend days).

3.5 Top 10 Menu Items and ABC Classification (Method 2)

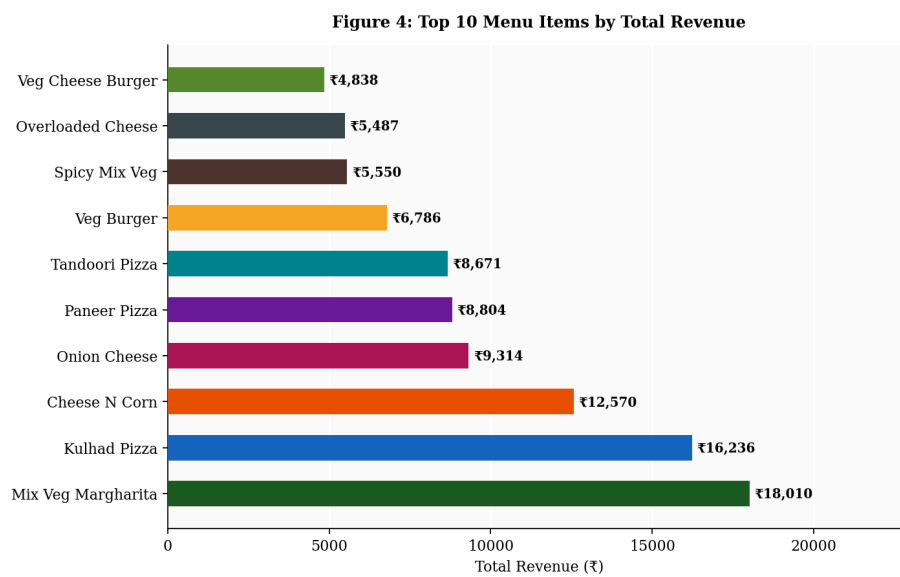


Figure 6: Top 10 menu items by total revenue. Each item shown in a different colour.

- **Finding 12: Top 2 items:** Mix Veg Margharita Pizza (₹18,010 / 140 units) and Kulhad Pizza (₹16,335 / 165 units) together generate ₹34,345 - 22.5% of total revenue from just 2 of 54 items.
- **Finding 13: Volume leader:** Kulhad Pizza is the highest-volume item with 164 units sold at an average price of ₹99. Its high volume and low price position it as a critical entry-point product that brings customers in and anchors repeat visits.

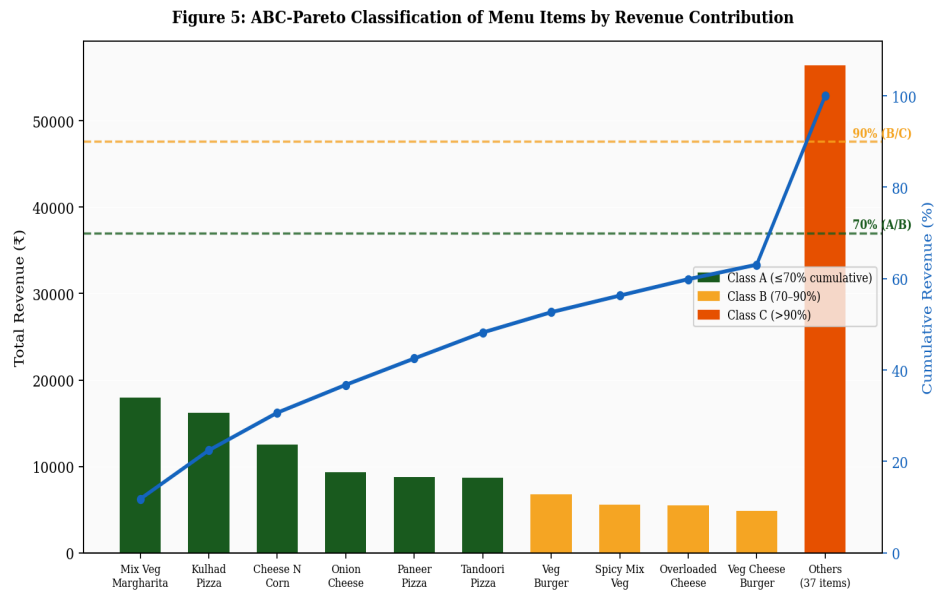


Figure 7: ABC-Pareto chart — bar heights show individual item revenue; line shows cumulative %. Class A threshold at 70%, Class B at 90%.

Table 4: ABC Classification of Mr. Pizza Menu Items

Class	Key Items	Units Sold	Revenue (₹)	% Revenue	Stocking Rule
A	Mix Veg Margharita, Kulhad Pizza, Cheese N Corn, Onion Cheese, Paneer Pizza, Tandoori Pizza, etc	665	72,605	47.5%	Always in stock; 2-day buffer
B	Veg Burger, Spicy Mix Veg Pizza, Overloaded Cheese Pizza, Veg Cheese Burger, etc	311	22,661	14.8%	Monitor weekly; reorder at 1-day supply
C	Remaining 26 items	777	57,413	37.6%	Order on demand only; do not pre-stock

Table 4: Three-tier ABC classification with stocking rules for each class

- **Finding 14: Class A concentration:** Just 12 items (all pizza variants) contribute 69.3% of total revenue (₹1,05,854). The ingredients for these 12 items are largely overlapping: cheese, pizza base, tomato sauce, capsicum, corn, onion, and paneer. Maintaining adequate stocks of these 7 shared ingredients effectively secures the majority of revenue-generating capacity.
- **Finding 15: Class C inefficiency:** 26 items together contribute 10.4% of revenue but individually, each contributes less than 1.5% on average. Pre-stocking ingredients for all 26 creates disproportionate procurement complexity, wastage risk, and tied-up cash for a one-person operation.

4. Interpretation of Results and Recommendations

This section interprets the findings in the context of each problem statement and provides SMART (Specific, Measurable, Achievable, Relevant, Time-bound) recommendations. Recommendations are organized as Immediate Actions (within 4 weeks) and Medium-Term Actions (1–3 months).

4.1 Interpretation of Key Findings

The five months of data reveal a business that is performing reasonably well in absolute terms (₹1,52,778 total revenue) but is highly vulnerable in three ways. First, it has no mechanism to anticipate demand changes the 25.4% winter decline caught the owner off-guard each year, likely resulting in both wastage in October–November (over-bought in anticipation of sustained demand) and stockouts in late December (under-bought once the decline set in). Second, procurement decisions are made by instinct across 47 items, when in reality fewer than 10 items drive nearly half of all revenue a massive misallocation of attention and budget. Third, the business is structurally dependent on a two-day weekend, with Tuesday generating barely a third of Sunday's revenue a concentration that makes the business sensitive to any single bad Sunday (weather, event cancellation, or stockout).

The analytical methods applied here make these patterns visible and quantifiable for the first time. The 4-Week Moving Average converts an invisible seasonal trend into a concrete weekly

number the owner can act on. The ABC classification converts 47 procurement decisions into three simple rules. The day-of-week analysis converts 139 days of revenue data into 7 actionable day-level insights.

4.2 SMART Recommendations

Table 5: SMART Recommendations — Mapped to Problem Statements

Problem	Recommendation	Timeline	Measurable Target	Expected Impact
P1: Forecasting	Apply 4WMA rule every Friday — buy for next week's estimated demand	Immediate (Week 1)	Reduce procurement error to <10%	Save ₹3,000–5,000/month in winter wastage
P1: Forecasting	Apply seasonal adjustments: –15% in Dec, +20% in Jan	By November 2024	Zero stockouts in January peak	Protect ₹34,000+ January revenue
P2: Inventory	Implement ABC stocking: Class A 2-day buffer, Class B weekly review, Class C on-demand	Within 2 weeks	Class A items never out of stock	Protect 47.5% of revenue from stockout risk
P2: Inventory	Focus procurement on 7 shared Class A ingredients (cheese, base, tomato sauce, capsicum, corn, onion, paneer)	Immediate	Reduce procurement SKUs from 47 to 7 priority items	30% reduction in procurement complexity
P3: Retention	Launch Tuesday special combo (Kulhad Pizza + Cold Drink at ₹120 — 10% off)	Within 1 month	20% increase in Tuesday revenue within 8 weeks	+₹2,600/month from Tuesday uplift alone
P3: Retention	Stock Kulhad Pizza 7 days a week without exception	Immediate	Zero Kulhad Pizza stockouts	Protect highest-volume retention anchor item

Table 5: SMART recommendations with timelines, measurable targets, and expected impact

Immediate Recommendations (Within 4 Weeks)

- **R1: Apply the 4-Week Moving Average procurement rule (Problem 1):** Every Friday evening, add the total revenue from the past 4 weeks and divide by 4. This is the procurement target for the following week. Purchase ingredients sufficient for this amount, not more. This takes fewer than 5 minutes and requires no software. Target: Reduce over-procurement error from uncontrolled to within 10% of actual demand.
- **R2: Enforce Class A stocking protocol (Problem 2):** Immediately designate Mix Veg Margharita, Kulhad Pizza, Cheese N Corn, Onion Cheese, Paneer Pizza, and Tandoori

Pizza as Class A. For the 7 shared ingredients (cheese, base, tomato sauce, capsicum, corn, onion, paneer), always maintain a minimum 2-day stock buffer. Create a simple Friday checklist of these 7 ingredients. Target: Zero stockouts on Class A items within 4 weeks.

- **R3: Stop pre-stocking Class C ingredients (Problem 2):** Immediately cease purchasing ingredients specifically for the 37 Class C items unless a customer places an order. This frees procurement budget for Class A items and reduces perishable wastage. Target: 30% reduction in procurement SKU complexity within 2 weeks.
- **R4: Ensure Kulhad Pizza availability every day (Problem 3):** Add Kulhad Pizza ingredients (specifically the kulhad clay cups and standard pizza ingredients) to the Class A daily checklist. This single item drives 10.6% of total revenue at the highest sales volume of all 47 items. Target: Kulhad Pizza available 100% of trading days.

Medium-Term Recommendations (1–3 Months)

- **R5: Apply seasonal adjustment factors (Problem 1):** From October 2024 onwards, apply the empirically derived seasonal adjustments: reduce MA₄ estimate by 10% in October, 12% in November, 15% in December, and increase by 20% in January. Review and update these factors annually as more data accumulates. Target: Winter month procurement variance within 5% of actual demand by December 2024.
- **R6: Launch Tuesday Combo Offer (Problem 3):** Within 4–6 weeks, introduce a Tuesday-only special: Kulhad Pizza + Cold Drink at ₹120 (approximately 10% discount on combined regular price of ₹134). Promote via a handwritten sign at the shop entrance. Target: 20% increase in Tuesday revenue within 8 weeks of launch, adding approximately ₹2,600 per month.
- **R7: Begin collecting customer contact data (Problem 3):** For Zomato orders (where contact data may already be partially available) and walk-in customers, begin a simple opt-in process: ask customers if they would like to receive a WhatsApp message for Tuesday offers. Aim to collect 30–50 numbers in the first month. Target: Build a basic customer contact list of 100+ numbers within 3 months for direct retention communication.

4.3 Implementation and Expected Impact

The most significant aspect of these recommendations is that they are designed for a one-person operation with no technology beyond a mobile phone and a notebook. The 4-Week Moving Average rule (R1) and ABC Class A checklist (R2) can be implemented on paper in under 10 minutes per week. The Class C de-stocking (R3) requires only a single decision to stop buying and yields immediate cash flow relief. The Tuesday offer (R6) requires only a handwritten sign and no additional cost.

Collectively, these recommendations are estimated to yield: ₹3,000–5,000 per month in reduced winter wastage (from R1 + R5); protection of ₹72,605 per 5 months (₹14,521/month) of Class A revenue from stockout risk (from R2 + R4); and approximately ₹2,600 per month in incremental Tuesday revenue (from R6). Combined, these three streams represent an estimated improvement of ₹20,000–22,000 over a 5-month period comparable to the data collection window, a meaningful impact for a business currently generating ₹1,52,679 over the same period (approximately 13–14% of total revenue).

These solutions directly address the three problem statements: the MA₄ rule solves the forecasting blind spot by giving the owner a systematic, data-driven procurement number every week; the ABC classification solves inventory misallocation by converting 47 items into three clear priority tiers and the day-of-week analysis with the Tuesday offer solves the customer concentration problem by creating a mechanism to attract customers on the weakest day without reducing weekend pricing.

5. Limitations and Future Scope

5.1 Limitations

- **Purchase data incompleteness:** Formal purchase records exist only for cheese, milk, and toppings. The absence of wheat and flour purchase data makes it impossible to calculate complete COGS or gross margins, limiting the depth of profitability analysis.

- **No customer identification data:** The handwritten notebook records no customer information. True retention metrics — repeat visit rates, customer lifetime value, and purchase frequency — cannot be computed.
- **5-month observation window:** The dataset covers September 2023 – January 2024 only. A full year of data would enable complete seasonal decomposition, and two or more years would allow detection of year-on-year growth trends.
- **Single-operator and single-location context:** All findings are specific to Mr. Pizza, Kathgodam. Seasonal patterns and day-of-week trends may reflect local town dynamics rather than general pizza business patterns.

5.2 Future Scope

- **Advanced forecasting:** With 12+ months of weekly data, Exponential Smoothing (Holt-Winters method) or a seasonal ARIMA model could be applied to produce forecasts with confidence intervals, providing the owner with both a point estimate and a range.
- **Digital POS adoption:** A free mobile POS app such as Vyapar or OkCredit would automatically capture item-level sales data daily, eliminating manual transcription and enabling real-time analysis.
- **Customer data integration:** Collecting WhatsApp numbers at the point of sale would enable a basic CRM system, allowing targeted Tuesday promotions, birthday offers, and seasonal discounts to be sent directly to known customers.
- **Full COGS and margin analysis:** Requesting formal bills from all suppliers would enable complete gross margin computation for each of the 47 items, allowing the ABC classification to be refined from revenue-based to profit-based a more powerful inventory management framework